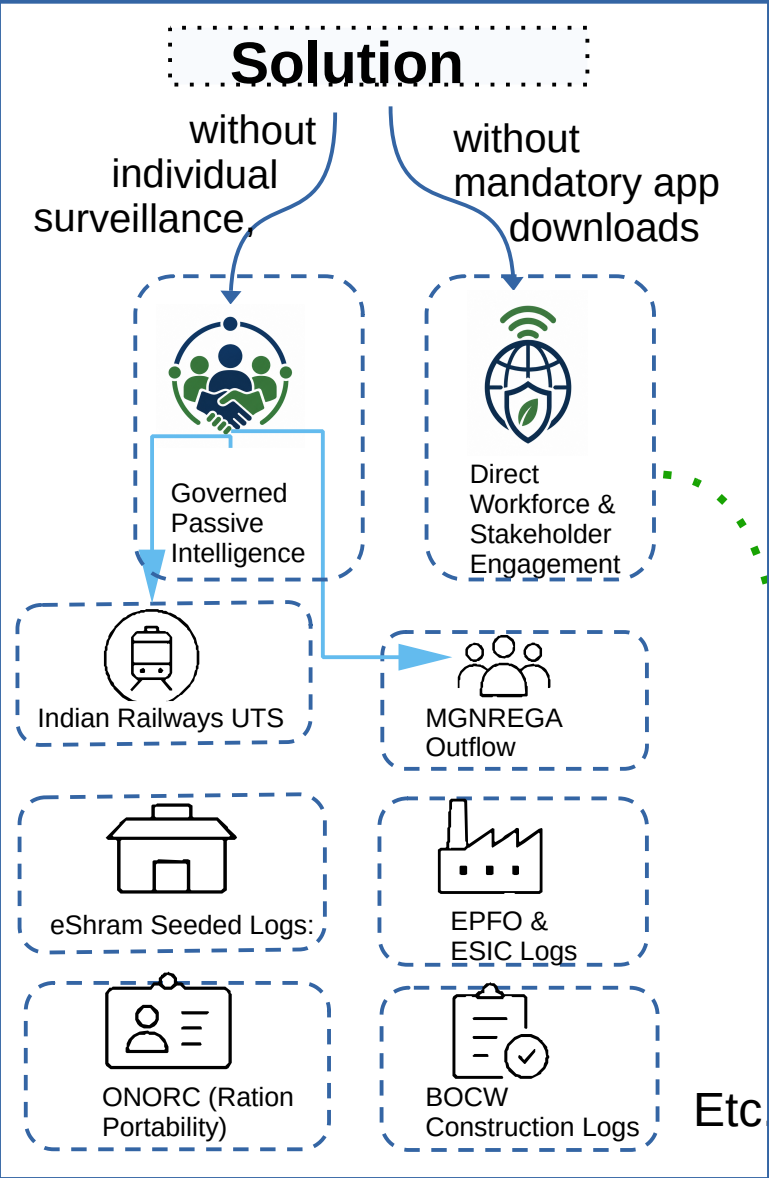
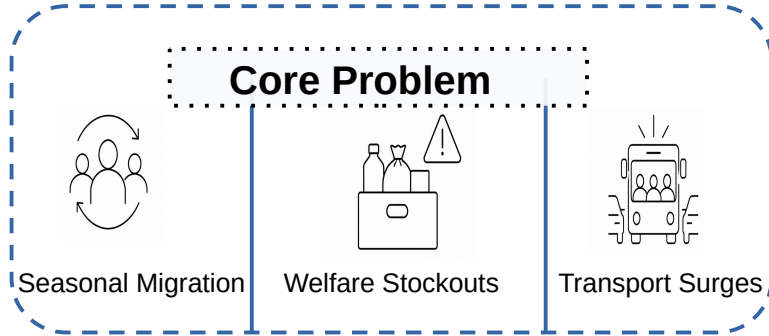


Shram - Predictive Mobility

Privacy-First Mass Migration Intelligence Engine



The Shram system is designed to ensure that seasonal and circular migrant workers face no administrative burden, mandatory app downloads, or pressure for manual check-ins. We capture baseline mobility using passive multi-source data integration while providing a dedicated direct-intake portal for enterprise and contractor-driven deployments.

CORE LAYER: High-Fidelity Direct Signals (Coverage Probability: ~65–75%)

- ONORC Ration Portability Logs
- Indian Railways Unreserved Ticketing (UTS) Volumes
- eShram Active Registration Seedings

Direct welfare/travel transaction - Governed Passive Intelligence

MIDDLE LAYER: Institutional & Statutory Signals (Coverage Probability: ~35–50%)

- MGNREGA Job Demand Drops / De-registrations
- EPFO & ESIC Monthly Registration Spikes
- BOCW Construction Site Welfare Logs
- State Inter-State Worker Registry Updates

Indirect statutory data — recorded by state government

OUTER LAYER: Macro Validation Signals (Coverage Probability: ~20–35%)

- Aggregated FASTag Toll Volumes
- Employer Bulk Dispatch Portal Data

Macro proxy — Aggregate infrastructure & transit volume data, individual

Coverage probability = estimated % of inter-state migrant movement captured via each signal layer. Figures based on published govt scheme enrollment data. To be validated in Phase 1 pilot.

Data References

1, 2, 3, 4, 5, 6

Tech : deep gravity,
References ,1,2,3,

Project github repo

Key Features

Federated Privacy

PII stays in state — always

Plug-in Architecture

New sources via adapters, no API lock-in

Low-Tech Worker

Channel Missed call / IVR / WhatsApp

Zero-Click Ingestion

Govt data, no worker burden

Policy Output

Role-Based Access Control

Secure, controlled data access

Causal Early Warning

Predictive alerts for emerging migration surges

Intelligence Layer

(Deep Gravity + Causal)

State Nodes

Local DB

DP Engine

Modular Adapter Design

Flexible integration with new data sources

Data Sources

Passive Feeds

Worker Opt-in

Deep Gravity Modeling

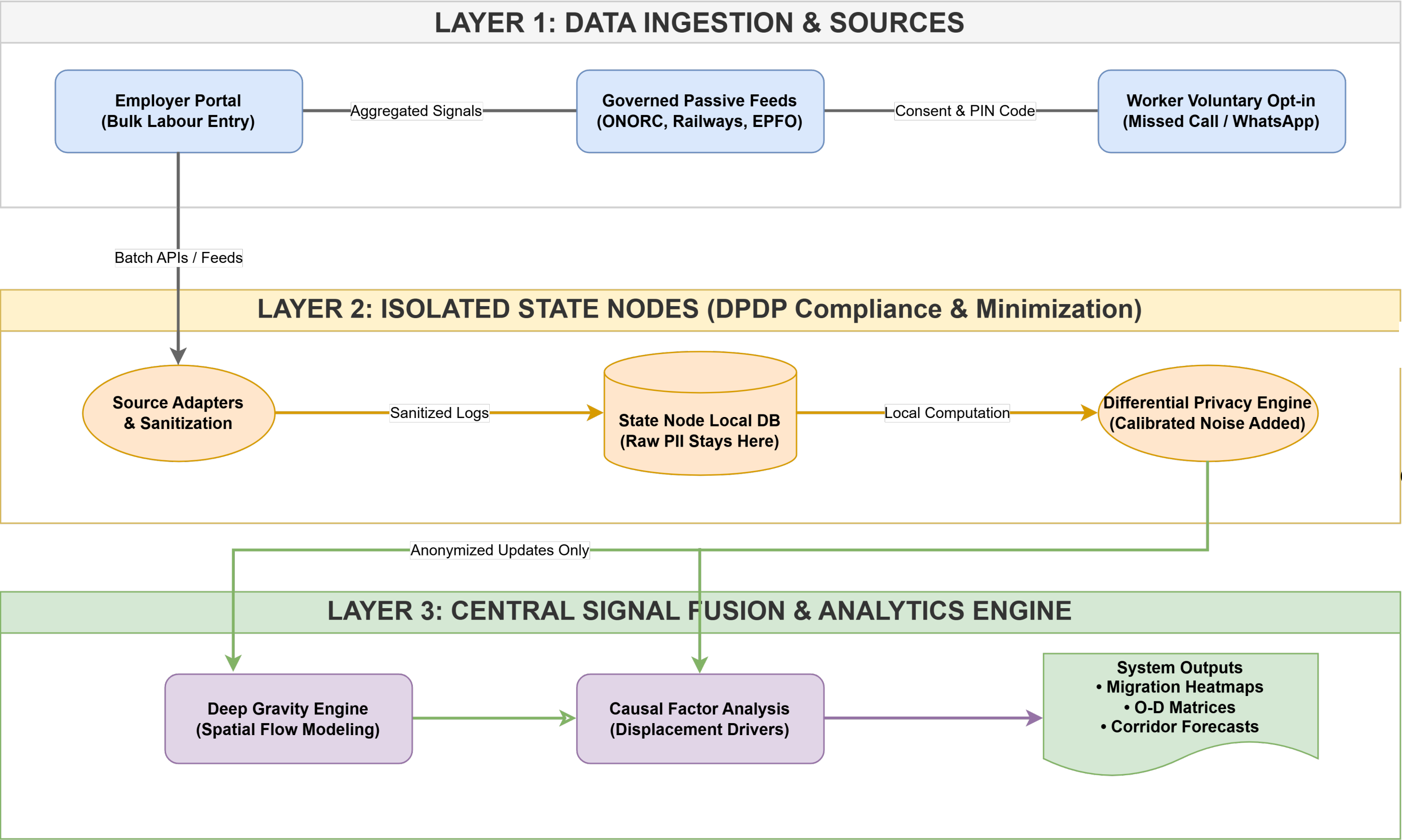
Non-linear flow prediction

Impact Targets

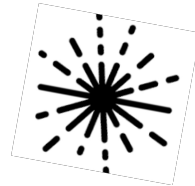
Mobility Flow Estimation = $\hat{F}_{ij}(t) = G \frac{P_i(t) P_j(t)}{d_{ij}^\beta} \cdot T$

Metric	Target
Welfare Stockout Reduction	20–25%
Predictive Lead Time	2–3 weeks
Privacy Guarantee	$\epsilon \leq 1.5, \delta \leq 10^{-5}$
Worker Channel Languages	22
Phase 1 Pilot Timeline	6–9 months
infrastructure	NIC/MeghRaj (no separate app cost)

ARCHITECTURE



Apache
kafka



Open Db



CausalNex
/ DoWhy

